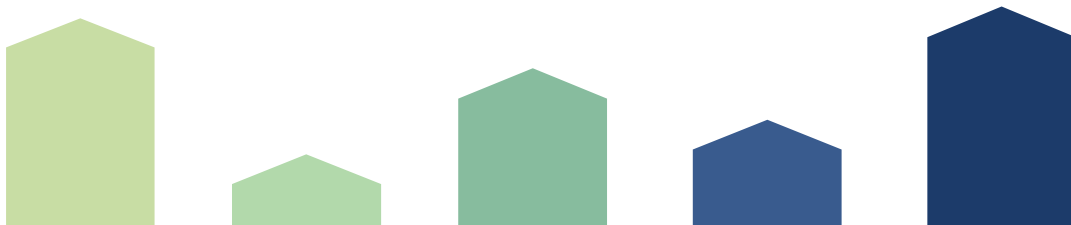


BSCS 506-Software Project Management

LECTURE#03

Project Management Methods and Tools



Learning Objectives:

- Understand various project management methods (Waterfall, Agile, SCRUM, Kanban, Lean, CPM)
- Learn about different project management tools and their applications (Gantt Charts, Trello, Jira, etc.)
- Identify suitable methods and tools for different types of projects
- Grasp the importance of communication, collaboration, and risk management tools

OUTLINE :

- Introduction to Software Project Management Methods and Tools,
- Project Management Methods,
 - Waterfall Model,
 - Agile Methodology,
 - SCRUM Framework,
 - Kanban Method,
 - Lean Method,
 - Critical Path Method (CPM)
- Key Project Management Tools,

Software Project Management Methods

- Software project management methods are systematic approaches used to plan, organize, and control the development of software projects.
- They provide a framework for managing the entire lifecycle of a project, from initial conception to final delivery.
These methods ensure that projects are completed on **time**, within **budget**, and to the desired **quality standards**.

2.1 Waterfall Methodology

- **Description:** A traditional, linear approach where tasks are completed in sequential phases.

- **Phases:**

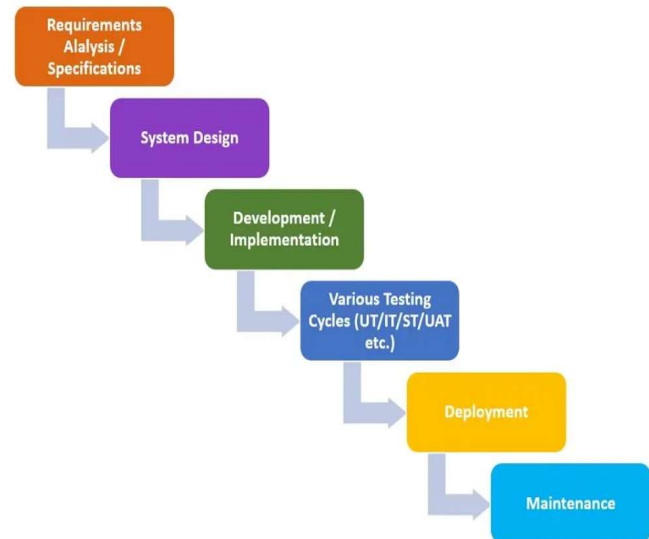
- Requirement gathering
- Design
- Development
- Testing
- Deployment

- **Advantages:**

- Clear structure with well-defined stages.
- Easier to manage for projects with fixed requirements.

- **Disadvantages:**

- Inflexible for changing requirements.
- Late testing phase can delay problem detection.



2.2 Agile Methodology

- **Description:** An iterative approach that emphasizes collaboration, flexibility, and customer feedback.

- **Core Principles:**

- Flexibility over strict adherence to plans.
- Close collaboration with the customer.
- Continuous delivery of working software.

- **Advantages:**

- High adaptability to changing requirements.
- Promotes collaboration and continuous improvement.

- **Disadvantages:**

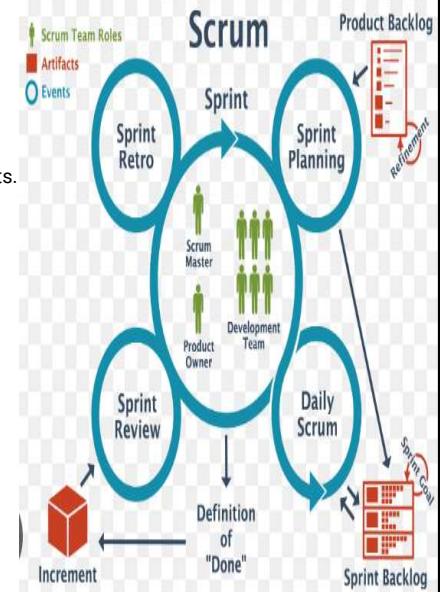
- Less predictable in terms of schedule and budget.
- Requires active customer involvement.



Fig. Agile Model

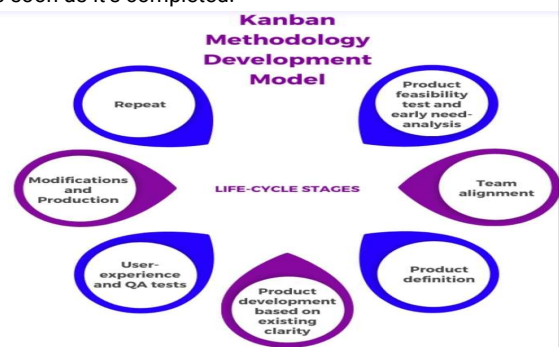
2.3 SCRUM

- **Description:** A popular Agile framework focusing on sprints (short, fixed-length iterations) to complete chunks of work.
- **Roles in SCRUM:**
 - **Product Owner:** Manages the product backlog and prioritizes work.
 - **SCRUM Master:** Facilitates the process and removes obstacles.
 - **Development Team:** Cross-functional team responsible for delivering increments.
- **Key Events:**
 - **Sprint:** Time-boxed iteration (usually 2-4 weeks).
 - **Sprint Planning, Daily Standups, Sprint Review, Sprint Retrospective.**
- **Advantages:**
 - Transparency and continuous feedback.
 - Incremental progress, allowing for early delivery of features.
- **Disadvantages:**
 - Needs disciplined teams and active collaboration.



2.4 Kanban

- **Description:** A visual workflow management system that helps teams visualize tasks, limit work in progress, and manage flow.
- **Key Elements:**
 - **Kanban Board:** Visual representation of the project workflow (tasks move through stages like "To Do," "In Progress," and "Done").
 - **Work In Progress (WIP) Limits:** Limit the number of tasks in progress to avoid overloading the team.
 - **Continuous Delivery:** No fixed iterations, work is delivered as soon as it's completed.
- **Advantages:**
 - Simple to implement and provides a clear visual overview.
 - Encourages focus on fewer tasks at a time.
- **Disadvantages:**
 - Less structured than SCRUM or Waterfall.
 - Harder to predict timelines for larger projects.



2.5 Lean Project Management

- **Description:** Focuses on delivering maximum value by eliminating waste and improving efficiency.

- **Principles:**

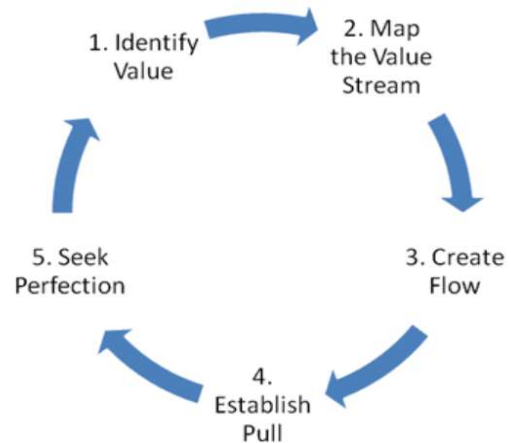
- Minimize waste (unnecessary processes, excessive meetings, over-engineering).
- Optimize workflows.
- Focus on customer value.

- **Advantages:**

- Reduces time and cost.
- Focuses on delivering the best value to the customer.

- **Disadvantages:**

- Requires deep understanding of what constitutes waste.
- Not suitable for projects with unclear customer needs.



2.6 Critical Path Method (CPM)

- **Description:** A technique for scheduling project activities based on identifying the longest path (critical path) through a project's task network.

- **Key Concepts:**

- **Critical Path:** The longest sequence of tasks that determines the minimum project duration.
- **Float:** The amount of time a task can be delayed without affecting the project end date.

- **Advantages:**

- Helps in identifying the most important tasks that directly affect the project timeline.
- Provides clear timelines for project completion.

- **Disadvantages:**

- May oversimplify complex projects.
- Does not handle resource allocation effectively.
- .

3. Project Management Tools

- **Software project management tools** are applications designed to assist in planning, organizing, and managing software development projects. They offer a range of features to streamline various aspects of the software development lifecycle (SDLC), such as task management, collaboration, version control, and project tracking.



3.1 Gantt Charts

- **Description:** A bar chart that visually represents the project schedule over time.
- **Features:**
 - Displays project tasks along with their start and end dates.
 - Helps in tracking progress and identifying potential delays.
- **Popular Tools:** Microsoft Project, Smartsheet.
- **Advantages:**
 - Easy to visualize timelines and dependencies between tasks.
 - Good for tracking overall project progress.
- **Disadvantages:**
 - Can become too complex for large projects.

3.2 Project Management Software : **Popular Tools:**

- **Trello:** A simple, Kanban-based project management tool.
- **Jira:** Popular for Agile and SCRUM projects, supporting sprint planning and backlog management.
- **Asana:** Task and project tracking with calendar views and task assignments.
- **Microsoft Project:** Advanced project scheduling with Gantt charts, resource allocation, and budgeting.
- **Monday.com:** Visual and easy-to-use project management tool with timeline and workload tracking.

Advantages:

- Centralized platform for task management, resource allocation, communication, and collaboration.
- Integration with other tools like Slack, Google Calendar, and GitHub.

Disadvantages:

- Can be expensive for large teams.
- Requires training for effective use.

3.3 Communication Tools

- **Description:** Tools that facilitate collaboration and communication between project team members, especially in remote and hybrid environments.
- **Popular Tools:**
 - **Slack:** A messaging platform with channels, direct messages, and integrations.
 - **Microsoft Teams:** A collaboration hub that integrates with Microsoft 365 for meetings, file sharing, and chat.
 - **Zoom:** Video conferencing tool for team meetings and client discussions.
- **Advantages:**
 - Improve collaboration and communication efficiency.
 - Enables remote work and real-time discussions.
- **Disadvantages:**
 - Too many channels can lead to scattered communication.

3.4 Time Management Tools

- **Description:** Tools that help track time spent on tasks, ensuring efficient time allocation and monitoring project progress.
- **Popular Tools:**
 - **Toggl:** Simple time-tracking tool with reports.
 - **Harvest:** Tracks time, expenses, and budgets.
 - **RescueTime:** Tracks how time is spent on devices to analyze productivity.
- **Advantages:**
 - Help in monitoring project timelines and deadlines.
 - Provides insights into team productivity and task bottlenecks.
- **Disadvantages:**
 - Requires discipline and consistency from team members.

3.5 Risk Management Tools

- **Description:** Tools designed to help identify, track, and mitigate risks throughout a project's lifecycle.
- **Popular Tools:**
 - **RiskWatch:** A tool for tracking and managing project risks.
 - **Active Risk Manager (ARM):** Risk management software that integrates with project schedules.
 - **RiskyProject:** Software for risk analysis and scheduling.
- **Advantages:**
 - Helps in early identification of risks.
 - Provides reports for proactive risk mitigation strategies.
- **Disadvantages:**
 - Can add complexity to project management.

3.6 Version Control Systems (for Software Projects)

- **Description:** Tools that help manage changes to software code and documents over time.
- **Popular Tools:**
 - **Git/GitHub:** Version control system and hosting service for collaborative software development.
 - **Bitbucket:** Similar to GitHub, but often used for private repositories.
 - **GitLab:** Comprehensive DevOps platform with version control and CI/CD pipelines.
- **Advantages:**
 - Tracks changes, enabling rollbacks to previous versions.
 - Facilitates collaboration among distributed teams.
- **Disadvantages:**
 - Requires familiarity with version control concepts.

4. Comparison of Methods and Tools

- **Waterfall vs. Agile:** Waterfall is more structured, Agile is more flexible.
- **SCRUM vs. Kanban:** SCRUM emphasizes time-boxed sprints, while Kanban focuses on limiting work in progress.
- **Choosing the Right Tools:** Based on project size, team structure, budget, and type of project (e.g., software, construction).

These tables provide a comprehensive overview of the **project management methods** and the **tools** that support them, helping students understand the practical application and benefits of each approach and technology.

Table 1: Project Management Methods

Method	Description	Phases	Advantages	Disadvantages	Best For
Waterfall	Linear, sequential model	Requirements, Design, Implementation, Testing, Deployment	Clear structure, easy to manage fixed requirements	Inflexible to changes, late problem detection	Projects with stable requirements
Agile	Iterative approach, prioritizes collaboration and flexibility	Iterations/Sprints, Continuous Feedback	Adapts to change, customer collaboration	Less predictability, requires active customer involvement	Dynamic projects with changing requirements
SCRUM	Agile framework focusing on time-boxed iterations (sprints)	Sprint Planning, Execution, Review, Retrospective	Transparency, early delivery of features, continuous feedback	Requires disciplined teams, less suitable for rigid projects	Teams that work well with frequent iterations
Kanban	Visual workflow system focused on managing task flow	Continuous task flow, no specific phases	Easy to implement, good for managing WIP (Work in Progress)	Lacks structured phases, harder to track progress for large projects	Continuous workflow projects (e.g., maintenance)
Lean	Focuses on eliminating waste and improving efficiency	Identify value, streamline workflows, optimize processes	Maximizes value, minimizes waste, reduces time and cost	May be difficult to implement in complex projects	Projects that aim to reduce waste and improve efficiency
Critical Path Method (CPM)	Identifies the longest path of dependent tasks to manage project duration	Task sequencing, time estimation, critical path analysis	Clear identification of key tasks, helps manage time	Not suitable for resource management, may oversimplify complex projects	Large projects with fixed timelines

Table 2: Key Project Management Tools

Tool	Description	Key Features	Advantages	Disadvantages	Best For
Gantt Charts	Visual bar chart showing project timeline and task durations	Task scheduling, start/end dates, dependency tracking	Easy visualization of timelines, tracks project progress	Can become complex for large projects	Time-bound projects that need task tracking
Trello	Simple, Kanban-based project management tool	Boards, lists, cards for task management	Easy-to-use, visual task tracking	Lacks advanced features for large teams or complex projects	Small to medium-sized projects
Jira	Agile project management tool, popular for SCRUM/Agile teams	Sprint planning, backlog management, reporting	Excellent for Agile teams, integrates with development tools	Can be overwhelming for non-technical teams	Agile software development projects
Asana	Task and project tracking tool with timeline/calendar views	Task assignments, progress tracking, 	Great for task management and tracking project	Limited advanced features for complex project	General-purpose project management

Table 2: Key Project Management Tools

Tool	Description	Key Features	Advantages	Disadvantages	Best For
Microsoft Project	Comprehensive project management software for Gantt charts, resource management	Resource allocation, timeline management, cost tracking	Robust project management features	Steep learning curve, expensive	Large projects needing resource and time tracking
Monday.com	Visual project management tool with custom workflows	Task boards, timelines, resource tracking	Highly customizable, good for managing team workload	Limited reporting features for large-scale analysis	Small to medium-sized projects
Slack	Communication platform for teams	Channels, direct messaging, file sharing	Real-time communication, integrates with project tools	Can lead to fragmented communication with too many channels	Teams needing real-time communication
Microsoft Teams	Collaboration hub for meetings, chat, file sharing	Video calls, chat, file sharing, app integration	Good for remote/hybrid work, integrates with MS Office tools	Can be overwhelming due to too many features	Corporate and remote team collaboration

Table 2: Key Project Management Tools

Tool	Description	Key Features	Advantages	Disadvantages	Best For
Zoom	Video conferencing tool	Video calls, screen sharing, breakout rooms	Easy to use, great for remote meetings	Limited project management capabilities	Remote meetings, team discussions
Toggl	Time tracking tool	Time tracking, reporting, project time allocation	Simple interface, provides insights into time spent on tasks	Can be too basic for advanced project needs	Freelancers, small teams needing time tracking
Harvest	Time and expense tracking tool	Time tracking, invoicing, budget management	Excellent for managing time and expenses	Requires discipline in manual time entry	Time-bound projects with financial tracking needs
Git/GitHub	Version control and collaboration tool for software projects	Code versioning, pull requests, issue tracking	Tracks changes, facilitates collaboration	Requires technical knowledge	Software development projects

Table 3: Methods vs. Tools for Project Management

Method	Best Supporting Tools	Why These Tools Are Suitable
Waterfall	Microsoft Project, Gantt Charts, Asana	These tools allow detailed task planning, timelines, and resource management.
Agile	Jira, Trello, Monday.com, Asana	Agile-focused tools emphasize iterations, sprint planning, and continuous delivery.
SCRUM	Jira, Trello, Slack, Microsoft Teams	Ideal for managing product backlogs, sprint planning, and facilitating daily SCRUM meetings.
Kanban	Trello, Monday.com	These tools provide visual task management using boards, lists, and cards, which align with Kanban's WIP-focused workflow.
Lean	Microsoft Project, Monday.com, Harvest	Time and resource tracking tools that support lean principles of efficiency and waste reduction.
Critical Path Method	Microsoft Project, Gantt Charts	Tools like Microsoft Project are suited for scheduling and critical path analysis, ensuring on-time project delivery.

Table 4: Comparison of Communication and Collaboration Tools

Tool	Primary Function	Features	Best For	Limitations
Slack	Team communication and collaboration	Channels, direct messaging, integrations	Small to large teams needing real-time communication	Can create scattered communication without discipline
Microsoft Teams	Team communication, meetings, file sharing	Video calls, chat, MS Office integration	Corporate teams, remote collaboration	Complex interface with too many features for small teams
Zoom	Video conferencing	Video calls, screen sharing, breakout rooms	Remote meetings, presentations	Limited in project management capabilities
GitHub	Version control and collaboration on software projects	Code versioning, issue tracking, pull requests	Distributed software development teams	Requires familiarity with version control systems

Table 5: Time and Risk Management Tools

Tool	Function	Best For	Advantages	Disadvantages
Toggl	Time tracking for tasks	Freelancers, small teams	Simple interface, easy time tracking	Limited for large-scale project management
Harvest	Time and expense tracking	Projects with budgets and financial tracking	Tracks both time and expenses, integrates with invoicing	Requires manual entry, can be complex for non-financial teams
RiskWatch	Risk management and tracking	Large projects with high risks	Helps identify, track, and manage risks	Over-complication for smaller, less risky projects
Active Risk Manager (ARM)	Risk analysis and project management	Projects with high uncertainty or risk	Integrates risk management into overall project planning	Expensive and complex to implement in simple projects

THANK YOU

